

From Wiersberg (1985) Applied Linear  
Regression

## APPENDIX

### 1A.1 A formal development of the simple regression model

Suppose we have two quantities, a predictor  $X$  and a response  $Y$ , and the actual relationship between  $X$  and  $Y$  is determined by some unknown function  $f$ , so that

$$Y = f(X) \quad (1A.1)$$

By collecting data, we want to study  $f$ , and thus study the relationship between  $X$  and  $Y$ . To this end, for each of  $n$  units or cases, we observe values  $x_i$  of  $X$  and  $y_i$  of  $Y$ ,  $i = 1, 2, \dots, n$ , where

$$y_i = f(x_i) + \varepsilon_i \quad (1A.2)$$

and  $\varepsilon_i$  is a random error representing variability in the observational process due to measurement error, neglected factors, and the like.

Now suppose that the shape of the unknown  $f$  can be approximated by a straight line. For this to be valid, the scales of  $X$  and/or  $Y$  may need to be changed or else the range of values of  $X$  considered may be limited. In any case, view the straight line as a first try at approximating  $f$ , and, if later analysis suggests this model to be inadequate, other analyses should be substituted. Thus  $f(x)$  is approximated by  $\beta_0 + \beta_1 x_i$  for some  $\beta_0$ ,  $\beta_1$  and

$$f(x_i) = \beta_0 + \beta_1 x_i + \delta_i \quad (1A.3)$$

where  $\delta_i$  is the fixed or lack of fit error reflecting the inadequacy of the straight line in matching  $f$ ,  $\delta_i = f(x_i) - \beta_0 - \beta_1 x_i$ . For the simple regression model to be useful, the  $\delta_i$  should be small (negligible) when compared to the  $\varepsilon_i$ .

Combining (1A.1) and (1A.2) gives the regression model:

where the  $\varepsilon_i$ 's, a component.

In this development, including errors where possible, it is useful to check the model.

### 1A.2 Measurement error

Suppose we let  $u_i$  be  $n+1$  known values of  $Y$ .

**E notation.** The variable  $u_i$ . The or less formally statement  $E(u_i)$  is the expected value of  $u_i$  that we observe.

The expectation of  $u_i$  is expressed by the  $E(u_i)$ .

For example, suppose  $E(u_i) = \mu$ , a constant of the sample  $(1/n)u_n$ , can be  $a_0 = 0$ . Thus

so that the same

**var Notation** variance is defined as

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Combining (1A.2) and (1A.3), and defining  $e_i = \varepsilon_i + \delta_i$ , we get the simple regression model

$$y_i = \beta_0 + \beta_1 x_i + e_i \quad i = 1, 2, \dots, n \quad (1A.4)$$

where the  $e_i$ 's, as in the text, consist of a fixed component and a random component.

In this development, we have taken the  $x_i$ 's to be measured without error. Including errors in the  $X$ 's complicates some of the analysis, and, whenever possible, it is useful to assume that errors in  $X$  are relatively small. Methodology for checking this assumption is a topic in Chapter 3.

## 1A.2 Means and variances of random variables

Suppose we let  $u_1, u_2, \dots, u_n$  be random variables, and also let  $a_0, a_1, \dots, a_n$  be  $n+1$  known constants.

***E* notation.** The symbol  $E(u_i)$  is read as the expected value of the random variable  $u_i$ . The term "expected value" is the same as the term "mean value," or less formally, the arithmetic average for a very large sample size. The statement  $E(u_i) = 0$  means that the average value we would get for  $u_i$ , if we sampled its distribution repeatedly, is 0; however, any specific realization of  $u_i$  that we observe is likely to be nonzero.

The expectation of a sum of random variables may be symbolically expressed by the following two equations:

$$E(a_0 + a_1 u_1) = a_0 + a_1 E(u_1) \quad (1A.5)$$

$$E\left(a_0 + \sum a_i u_i\right) = a_0 + \sum a_i E(u_i) \quad (1A.6)$$

For example, suppose that  $u_1, u_2, \dots, u_n$  make up a random sample, and  $E(u_i) = \mu$ , a constant value for all  $i = 1, 2, \dots, n$ . Then, the expected value of the sample average of the  $u_i$ 's,  $\bar{u} = \sum u_i / n = (1/n)u_1 + (1/n)u_2 + \dots + (1/n)u_n$ , can be found from equation (1A.6) with  $a_i = 1/n$ ,  $i = 1, 2, \dots, n$ , and  $a_0 = 0$ . Thus

$$E(\bar{u}) = \sum \left(\frac{1}{n}\right) E(u_i) = \left(\frac{1}{n}\right) \sum \mu = \left(\frac{1}{n}\right) n\mu = \mu \quad (1A.7)$$

so that the sample average is an unbiased estimate of the mean  $\mu$ .

***var* Notation.** The symbol  $\text{var}(u_i)$  is read as the variance of  $u_i$ . The variance is defined by the equation  $\text{var}(u_i) = E[u_i - E(u_i)]^2 =$  the expected

square of the difference between an observed value for  $u_i$  and its average value. The larger  $\text{var}(u_i)$  is, the more variable observed values for  $u_i$  are likely to be. The symbol  $\sigma^2$  is often used for a variance, or  $\sigma_u^2$  might be used for the variance of the  $u$ 's when several variances are being discussed.

The general rule for the variance of a sum of random variables (if the variables are uncorrelated) is

$$\text{var}\left(a_0 + \sum a_i u_i\right) = \sum a_i^2 \text{var}(u_i) \quad (1A.8)$$

The  $a_0$  term vanishes: the variance of a constant is zero. We can now apply this equation to find the variance of the sample average, assuming that the  $u_i$ 's are uncorrelated with common variance  $\text{var}(u_i) = \sigma_u^2$ :

$$\text{var}\left(\sum \left(\frac{1}{n}\right) u_i\right) = \sum \left(\frac{1}{n}\right)^2 \text{var}(u_i) = n \left(\frac{1}{n}\right)^2 \sigma_u^2 = \frac{\sigma_u^2}{n}$$

**cov Notation.** The symbol  $\text{cov}(u_i, u_j)$  is read to be the covariance between the random variables  $u_i$  and  $u_j$  and is defined by the equation  $\text{cov}(u_i, u_j) = E[(u_i - E(u_i))(u_j - E(u_j))]$ . The covariance describes the way two random variables jointly vary. If the two variables are independent, then they are uncorrelated, but not necessarily conversely. If  $i=j$  in the definition, then we see that  $\text{cov}(u_i, u_i) = \text{var}(u_i)$  by the definition of the latter symbol. The rule for covariance is

$$\text{cov}(a_0 + a_1 u_1, a_3 + a_2 u_2) = a_1 a_2 \text{cov}(u_1, u_2) \quad (1A.9)$$

Often, a scale-free version of the covariance is used in its place. This is called the *correlation coefficient*, abbreviated as  $\text{corr}(u_i, u_j)$ , and is defined by the equation

$$\text{corr}(u_i, u_j) = \frac{\text{cov}(u_i, u_j)}{\sqrt{\text{var}(u_i)\text{var}(u_j)}} \quad (1A.10)$$

The correlation does not depend on the units of measurement for the random variables and has a value between  $+1$  and  $-1$ . If the correlation is zero, then the variables  $u_i$  and  $u_j$  are uncorrelated; this will happen only if  $\text{cov}(u_i, u_j) = 0$ .

The general form of the variance of linear combination of random variables depends both on the variances of the variables, and on their covariances, according to the following rule:

$$\text{var}(a_0 + \sum a_i u_i) = \sum_{i=1}^n a_i^2 \text{var}(u_i) + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n a_i a_j \text{cov}(u_i, u_j) \quad (1A.11)$$

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### 1A.4 Mean

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## 1A.3 Least squares

The least squares estimates of  $\beta_0$  and  $\beta_1$  in simple regression are those values  $\hat{\beta}_0$  and  $\hat{\beta}_1$  that minimize the residual sum of squares function,

$$RSS(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 \quad (1A.12)$$

One method of minimizing (1A.12) is to differentiate with respect to  $\beta_0$  and  $\beta_1$ , set the derivatives equal to zero, and solve the resulting equations. Carrying out this plan,

$$\begin{aligned} \frac{\partial RSS(\beta_0, \beta_1)}{\partial \beta_0} &= -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \\ \frac{\partial RSS(\beta_0, \beta_1)}{\partial \beta_1} &= -2 \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) = 0 \end{aligned} \quad (1A.13)$$

Upon rearranging terms, (1A.13) becomes

$$\begin{aligned} \beta_0 n + \beta_1 \sum x_i &= \sum y_i \\ \beta_0 \sum x_i + \beta_1 \sum x_i^2 &= \sum x_i y_i \end{aligned} \quad (1A.14)$$

Equations (1A.14) are called the *normal equations* for model (1.2). The data are used only through the aggregates or sufficient statistics  $\sum x_i$ ,  $\sum y_i$ ,  $\sum x_i^2$ , and  $\sum x_i y_i$ , or, equivalently, through the numerically more stable  $\bar{x}$ ,  $\bar{y}$ ,  $SXX$ , and  $SXY$ ; these are the aggregate statistics that would be found if the deviations from sample average form of the model were used. Any two sets of data with these quantities the same will have the same estimators  $\hat{\beta}_0$  and  $\hat{\beta}_1$ . Solving the two linear equations (1A.14) gives

$$\begin{aligned} \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} \\ \hat{\beta}_1 &= \frac{SXY}{SXX} \end{aligned} \quad (1A.15)$$

## 1A.4 Means and variances of least squares estimates

The least squares estimates are linear functions of the  $y_i$ ,  $i = 1, \dots, n$ , and, since the  $y_i$ 's are linear functions of the  $e_i$ 's, we will be able to apply the results of section 1A.2 to the estimates found in 1A.3 to get the means, variances, and covariances of the estimates. In particular, assume the simple regression

model

$$y_i = \beta_0 + \beta_1 x_i + e_i \quad i = 1, 2, \dots, n$$

is correct. By results (1A.6) and (1A.8),  $E(y_i) = \beta_0 + \beta_1 x_i$ , and  $\text{var}(y_i) = \text{var}(e_i) = \sigma^2$ . Now, consider the estimate  $\hat{\beta}_1$ , given in (1A.15). Suppose we define the constants  $c_1, c_2, \dots, c_n$  by the equation (for each  $i$ )

$$c_i = \frac{(x_i - \bar{x})}{SXX} \quad i = 1, 2, \dots, n$$

Since the  $x_i$  are considered to be fixed numbers, so are the  $c_i$ . The estimator  $\hat{\beta}_1$  is equal to  $\sum c_i y_i$ , a linear combination of the  $y_i$ 's. The mean of  $\hat{\beta}_1$  is then found as

$$\begin{aligned} E(\hat{\beta}_1) &= E\left(\sum c_i y_i\right) = \sum c_i E(y_i) \\ &= \sum c_i (\beta_0 + \beta_1 x_i) = \beta_0 \sum c_i + \beta_1 \sum c_i x_i \end{aligned}$$

But one can show by direct summation  $\sum c_i = 0$ ,  $\sum c_i x_i = 1$ , giving

$$E(\hat{\beta}_1) = \beta_1 \quad (1A.16)$$

which shows that, as long as  $E(y_i) = \beta_0 + \beta_1 x_i$ ,  $\hat{\beta}_1$  is an unbiased estimate of  $\beta_1$ . Also, one can easily show that  $E(\hat{\beta}_0) = \beta_0$ .

The variance of  $\hat{\beta}_1$  is found by

$$\begin{aligned} \text{var}(\hat{\beta}_1) &= \text{var} \sum (c_i y_i) \\ &= \sum c_i^2 \text{var}(y_i) + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n c_i c_j \text{cov}(y_i, y_j) \end{aligned}$$

but  $\text{cov}(y_i, y_j) = \text{cov}(\beta_0 + \beta_1 x_j + e_j, \beta_0 + \beta_1 x_i + e_i) = \text{cov}(e_j, e_i) = 0$  by assumption. Also  $\text{var}(y_i) = \text{var}(e_i) = \sigma^2$  by assumption. Hence

$$\text{var}(\hat{\beta}_1) = \sigma^2 \sum c_i^2$$

but  $\sum c_i^2 = 1/SXX$ , so that

$$\text{var}(\hat{\beta}_1) = \sigma^2 \frac{1}{SXX} \quad (1A.17)$$

To find the variance of  $\hat{\beta}_0$ , write

$$\begin{aligned} \text{var}(\hat{\beta}_0) &= \text{var}(\bar{y} - \hat{\beta}_1 \bar{x}) \\ &= \text{var}(\bar{y}) + \bar{x}^2 \text{var}(\hat{\beta}_1) - 2\bar{x} \text{cov}(\bar{y}, \hat{\beta}_1) \end{aligned}$$

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Now,  $\text{var}(\bar{y}) = \sigma^2/n$ ,  $\text{var}(\hat{\beta}_1)$  is given by (1A.17), and  $\text{cov}(\bar{y}, \hat{\beta}_1) = 0$ . This last result can be shown by application of the rules in the last section, but it is intuitively clear because the average value  $\bar{y}$  should not depend in any way on the fitted slope  $\hat{\beta}_1$ . Thus we get

$$\text{var}(\hat{\beta}_0) = \sigma^2 \left( \frac{1}{n} + \frac{\bar{x}^2}{SXX} \right) \quad (1A.18)$$

Finally,

$$\begin{aligned} \text{cov}(\hat{\beta}_0, \hat{\beta}_1) &= \text{cov}(\bar{y} - \hat{\beta}_1 \bar{x}, \hat{\beta}_1) \\ &= \text{cov}(\bar{y}, \hat{\beta}_1) - \bar{x} \text{cov}(\hat{\beta}_1, \hat{\beta}_1) \\ &= -\frac{\sigma^2 \bar{x}}{SXX} \end{aligned} \quad (1A.19)$$

A further application of these results gives the variance of a fitted value  $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ :

$$\begin{aligned} \text{var}(\hat{y}_i) &= \text{var}(\hat{\beta}_0 + \hat{\beta}_1 x_i) \\ &= \text{var}(\hat{\beta}_0) + x_i^2 \text{var}(\hat{\beta}_1) + 2x_i \text{cov}(\hat{\beta}_0, \hat{\beta}_1) \\ &= \sigma^2 \left( \frac{1}{n} + \frac{\bar{x}^2}{SXX} \right) + \sigma^2 x_i^2 \frac{1}{SXX} - 2\sigma^2 x_i \frac{\bar{x}}{SXX} \\ &= \sigma^2 \left[ \frac{1}{n} + \frac{(x_i - \bar{x})^2}{SXX} \right] \end{aligned} \quad (1A.20)$$

The variance of a prediction  $\tilde{y}_*$  of the future value  $y_*$  at  $x_*$  is a bit more complicated. Both  $\tilde{y}_*$  and  $y_*$  have the same expectation, so we can write

$$\begin{aligned} E(\tilde{y}_* - y_*)^2 &= E\{[\tilde{y}_* - E(y_*)] - [y_* - E(y_*)]\}^2 \\ &= E[\tilde{y}_* - E(y_*)]^2 + E[y_* - E(y_*)]^2 \\ &\quad - 2E[\tilde{y}_* - E(y_*)][y_* - E(y_*)] \end{aligned}$$

Since  $\tilde{y}_*$  is computed from the past data, and  $y_*$  is an observation in the future, the prediction and the future value are uncorrelated, and the covariance term in the last expression is zero. The first term in square brackets is the same as (1A.20), with  $x_i$  replaced by  $x_*$ . The second term in square brackets is just the variance of the future value, equal to  $\sigma^2$ . Thus the last expression is

$$\text{var}(\tilde{y}_*) = \sigma^2 \left[ \frac{1}{n} + \frac{(x_* - \bar{x})^2}{SXX} \right] + \sigma^2 \quad (1A.21)$$

which, upon rearranging terms, gives (1.36).